AoyuX Monthly Earthquake Forecasting Report for China Seismic Experimental Site (CSES)

version 1

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Monthly report for February 1, 2025 to February 28, 2025 (28 days)
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1. Earthquake Forecasts for the Next Month in the CSES

This report uses the earthquake catalog up to and including January 31, 2024, with a forecast time window covering the following month (i.e., from February 1 to February 28, 2025, a total of 28 days). The catalog uses local magnitude (M_L) as the measure of earthquake size. The local magnitudes are published by the China Earthquake Networks Center (CENC). The earthquake forecasts are generated using simulations based on the Epidemic Type Aftershock Sequence (ETAS) model with spatially varying background rates (Nandan et al., 2021a; 2021b, 2022; Li et al., 2024).

It is expected that, in the next month, the level of seismicity within the CSES (Chuan-Yunnan region) will slightly increase compared to the previous month (January 2025). The probabilities of at least one earthquake of magnitude 4 or larger, 5 or larger, and 6 or larger occurring in the region are 91%, 29% and 4%, respectively.

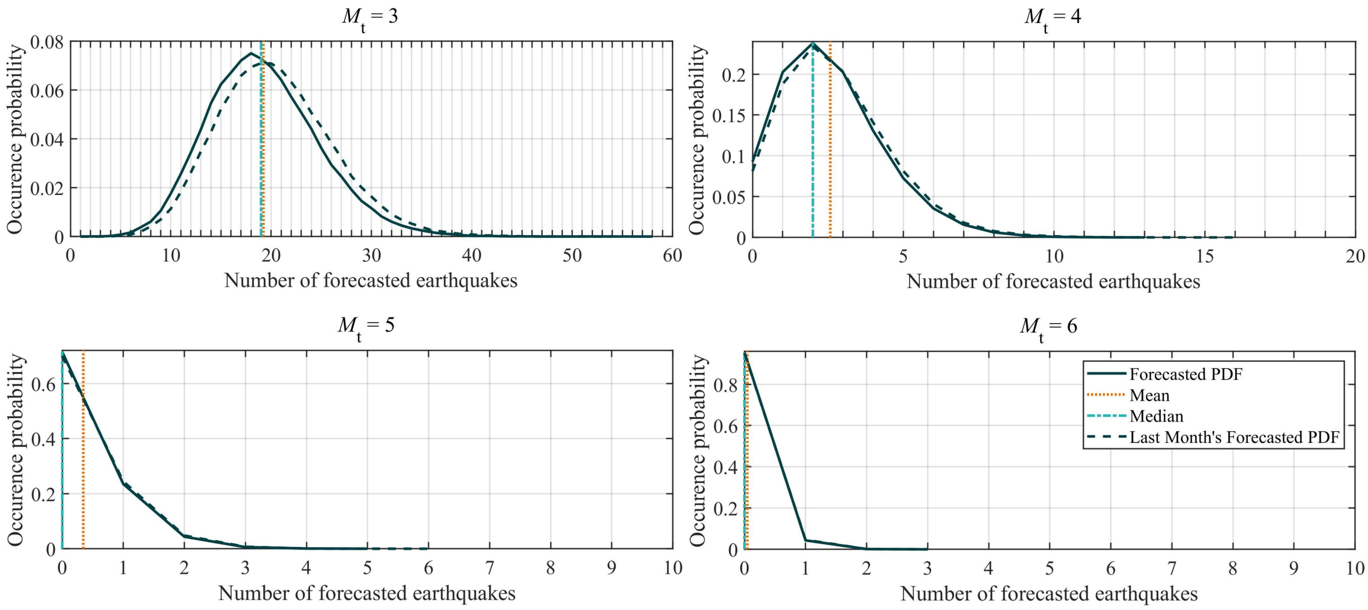


Figure 1. Forecasted probabilities of the numbers of earthquakes of different magnitude ranges ($M \geq M_t$) for the next month.

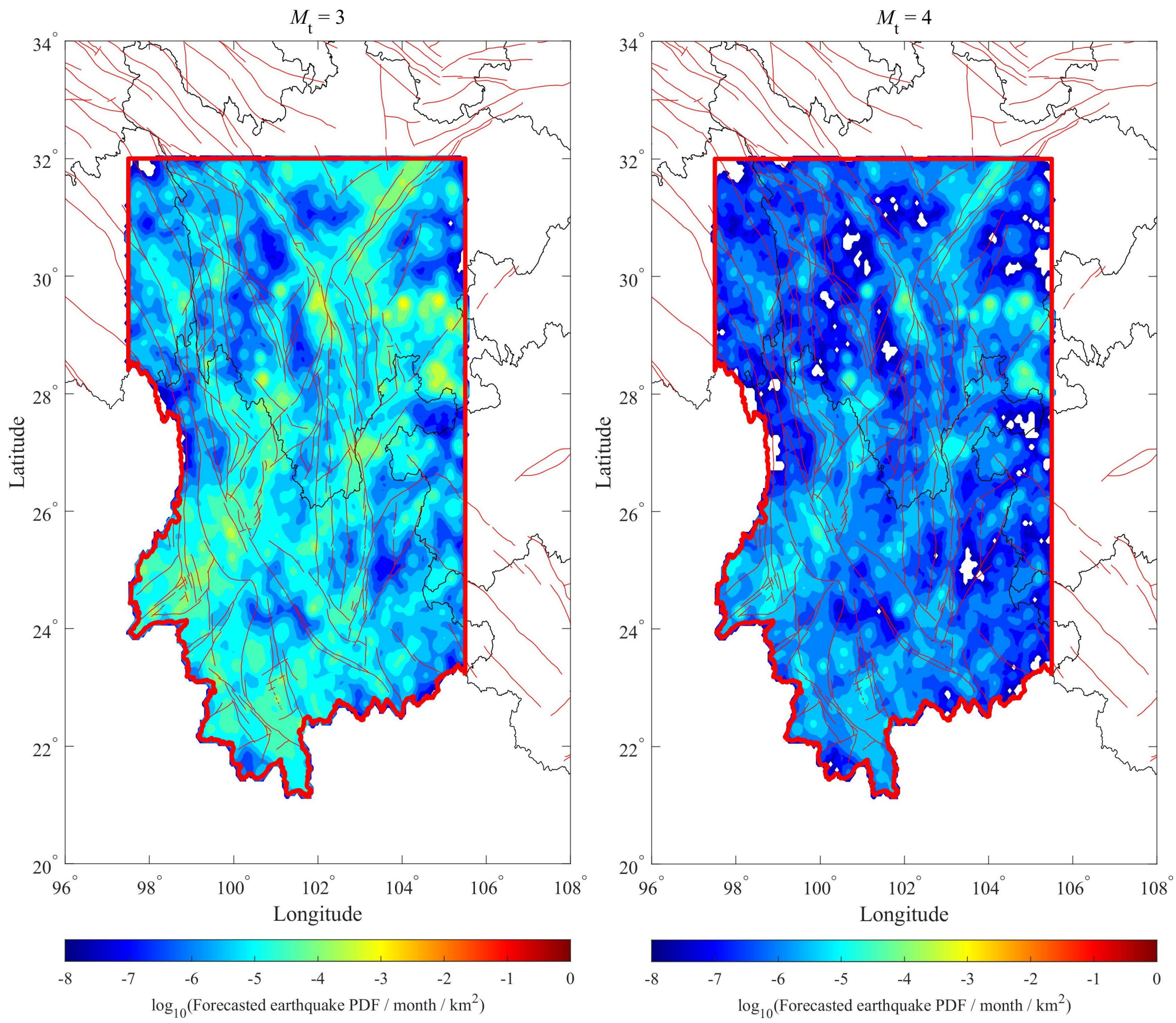


Figure 2. Probability density distribution (PDF) of the numbers of forecasted earthquakes ($M \geq M_t$) for the next month per unit area (km^2) and per month.

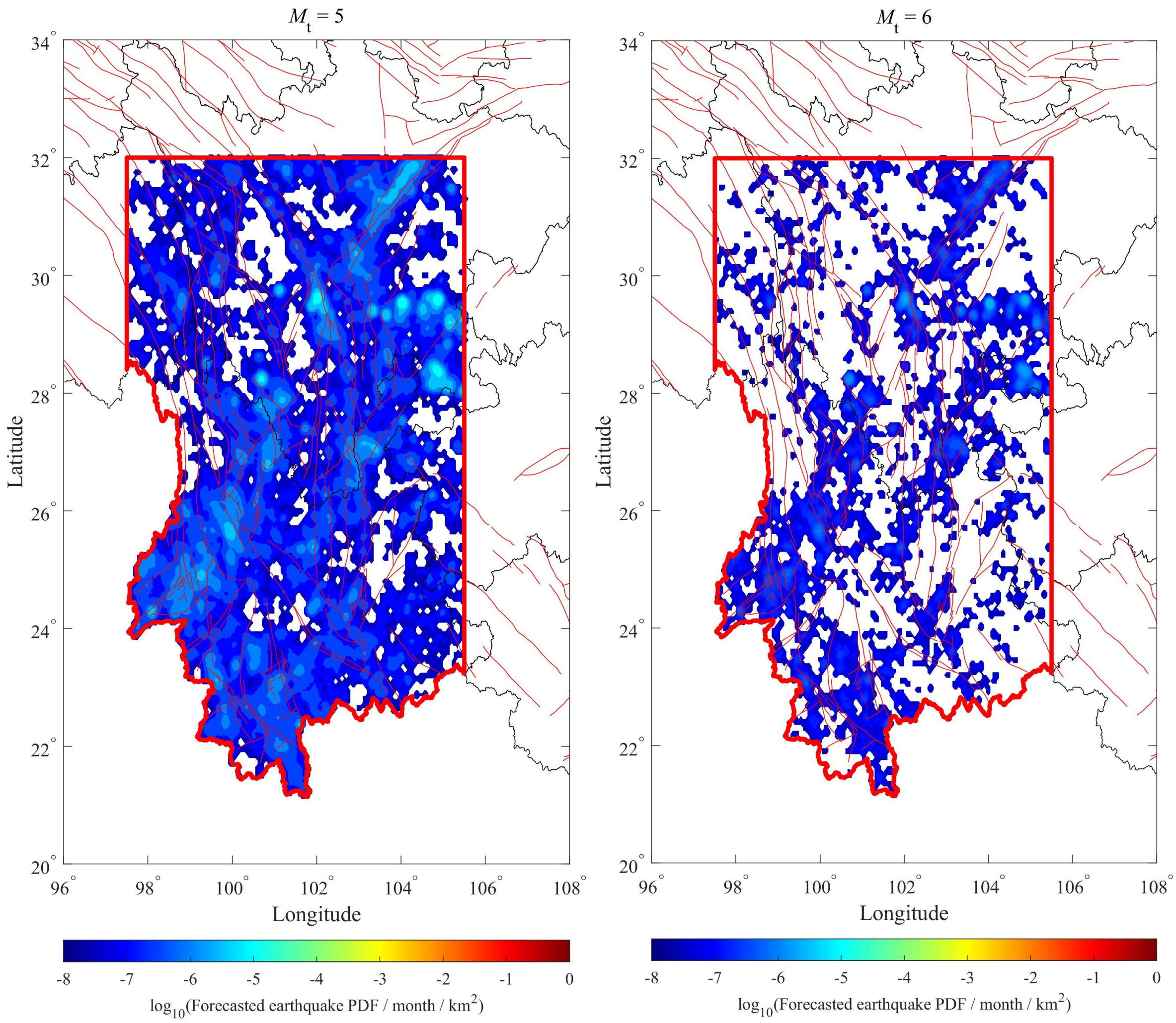


Figure 2. -continued.

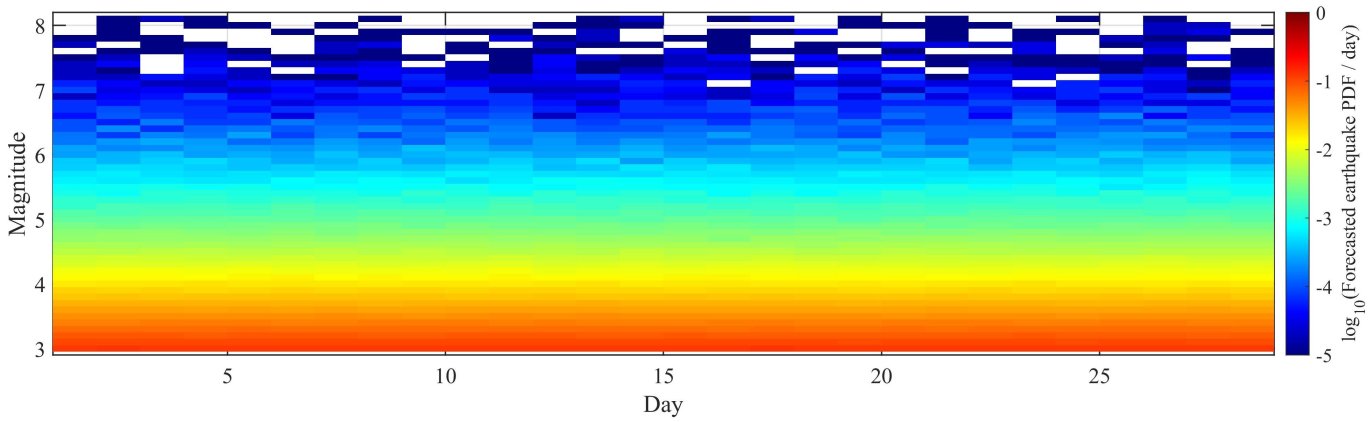


Figure 3. Daily probability density distribution (PDF) of the number of forecasted earthquakes per day over the whole CSES region for the next month.

2. Last Month's Forecast *versus* Realised Seismicity

In the last month (i.e., from January 1 to January 31, 2024, a total of 31 days), the region has experienced 24 earthquakes of M 3+, 5 earthquakes of M 4+, with no earthquakes of M 5+.

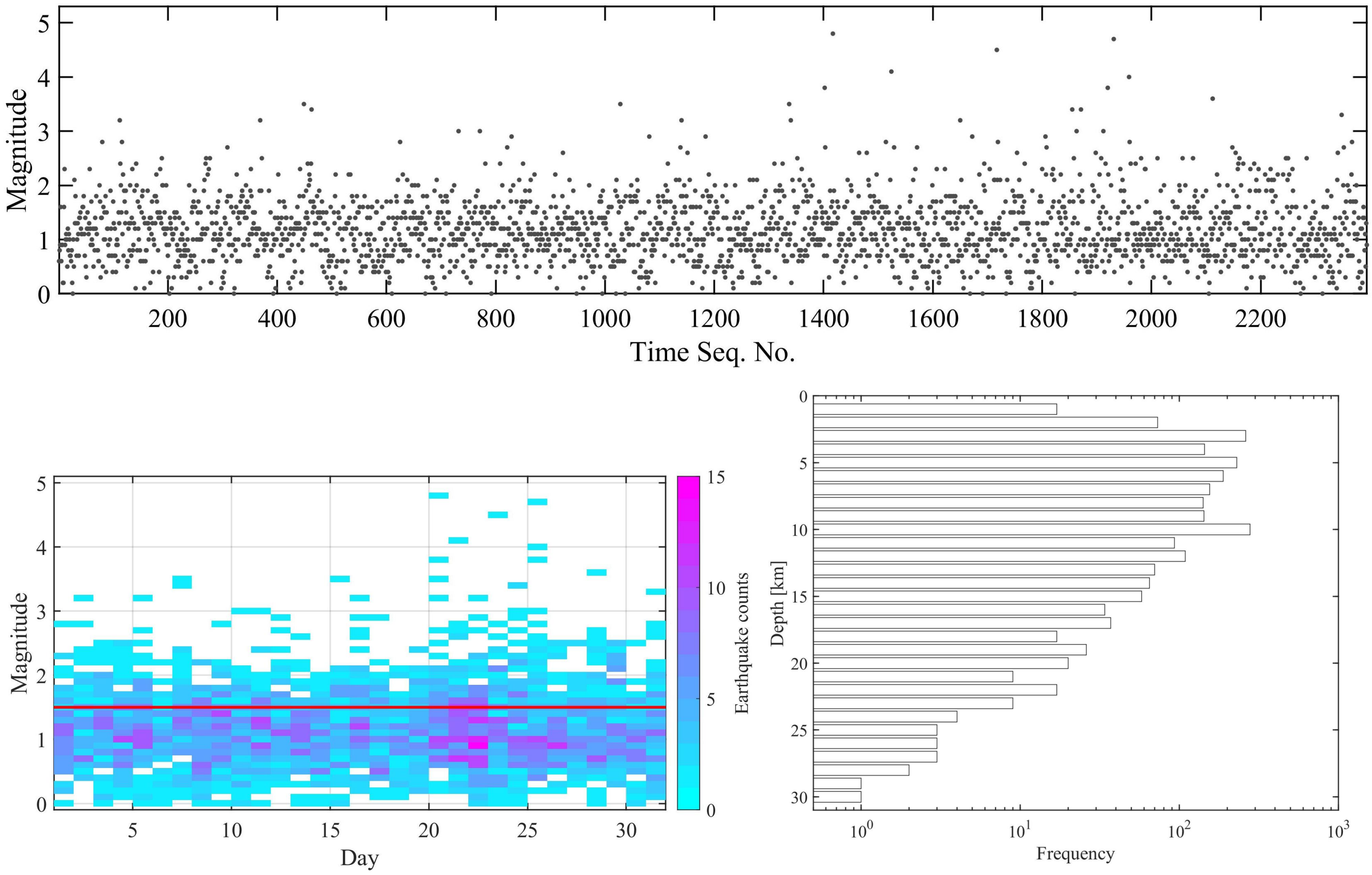


Figure 4. Seismicity in the CSES over the last month. (Top) Magnitude-time sequence plot; (Bottom left) Daily earthquake counts per 0.1 magnitude bin. (Bottom right) Depth-frequency distribution of earthquakes. The red line in the bottom-left figure represents the completeness magnitude (M_c) of the catalog from the last month.

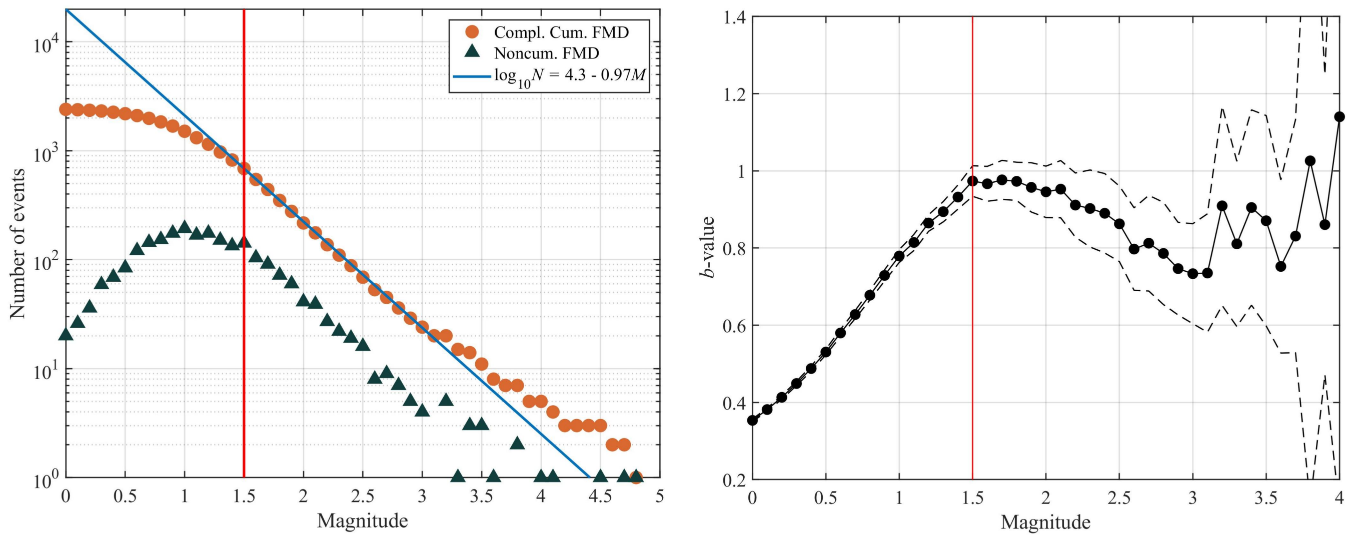


Figure 5. (left) Complementary cumulative and density frequency-magnitude distribution of seismicity from the last month in the CSES, alongside the Gutenberg-Richter (GR) law fitted using earthquakes with $M \geq M_c$, where M_c is derived for the last month (red vertical line); (right) Variation of the b-value of the GR law with M_{co}, with the thin dashed lines indicating the $\pm 1\sigma$ standard deviation boundaries.

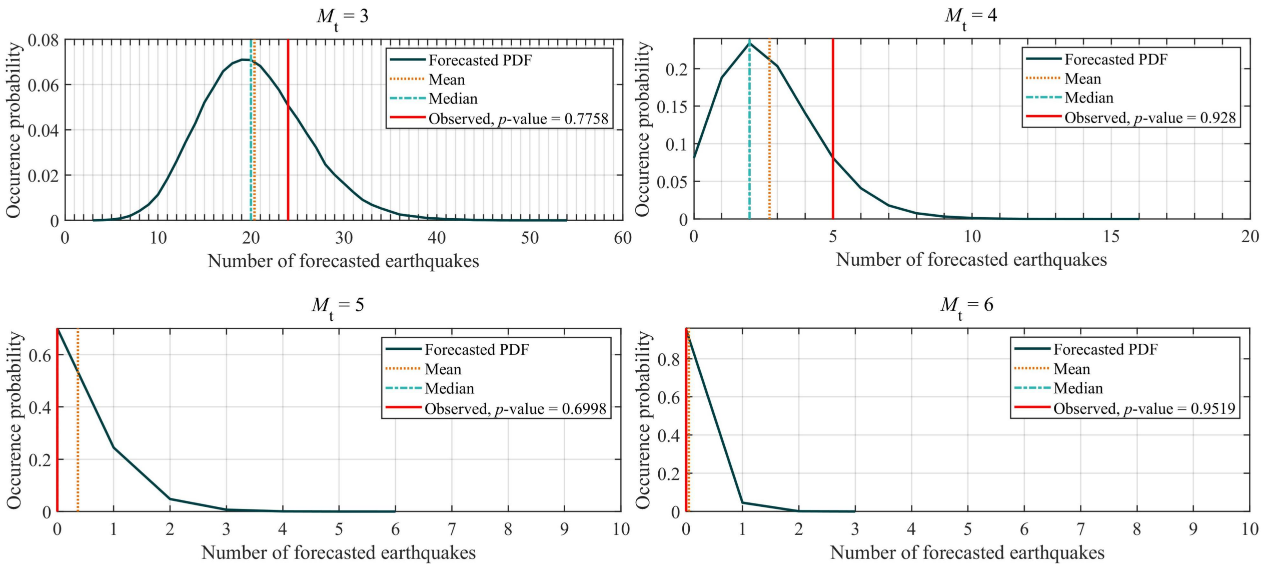
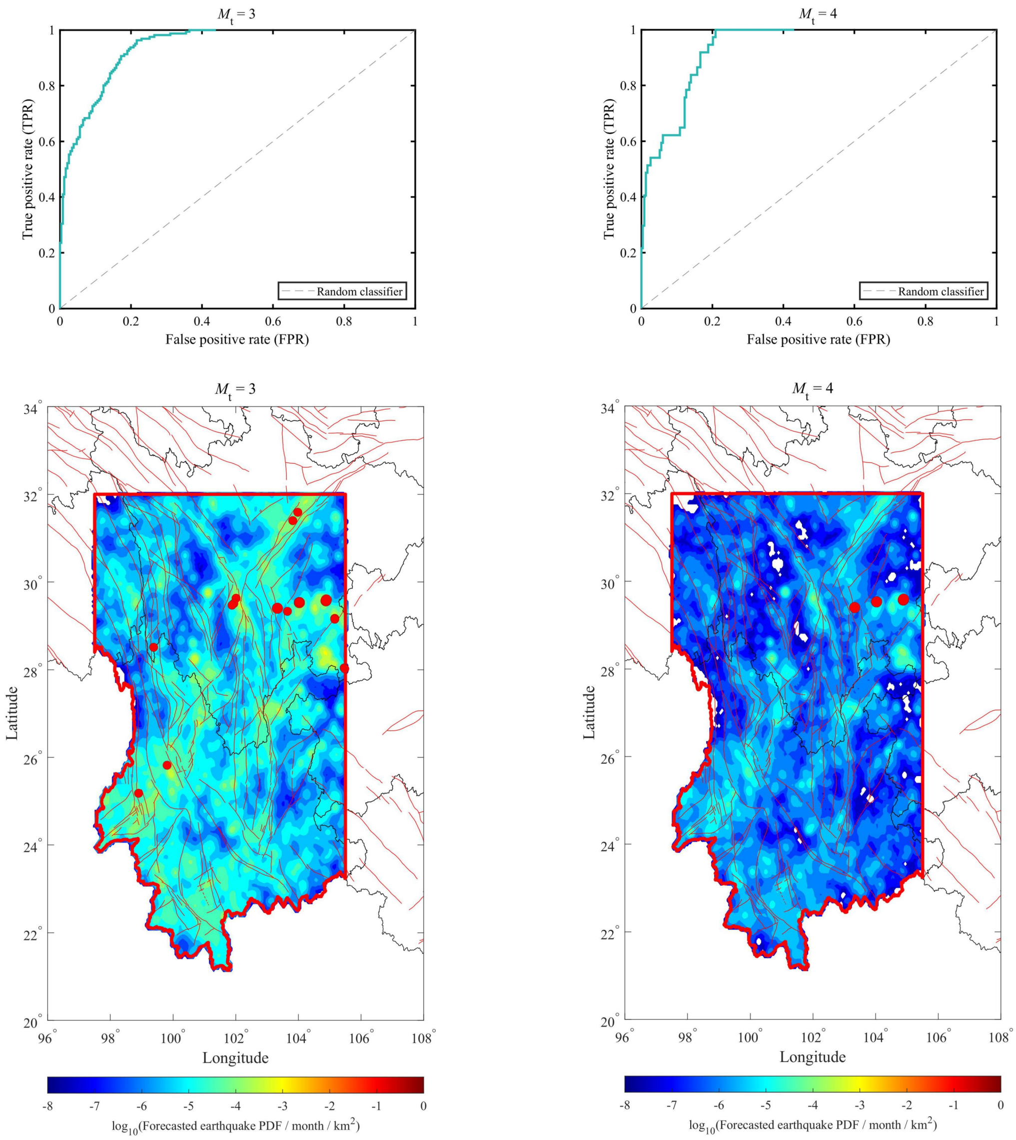


Figure 6. Number of observed earthquakes ($M \geq M_t$) in the CSES over the last month (red vertical line) and forecasted probabilities for earthquakes of different magnitude ranges ($M \geq M_t$) published last month. The p-value is the cumulative distribution function (CDF) value for the number of observed earthquakes.



No ROC curve results for $M_t = 5$;　　No ROC curve results for $M_t = 6$;

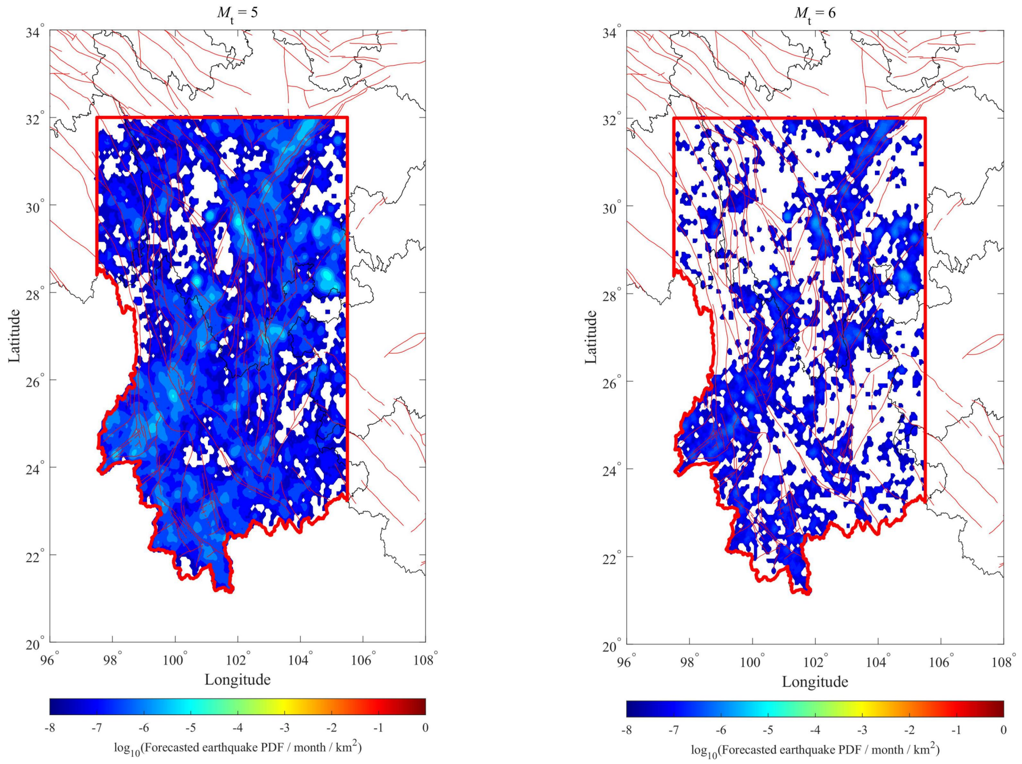


Figure 7. Top: Receiver Operating Characteristic (ROC) curve. The four following panels show the probability density distribution (PDF) of forecasted earthquakes ($M \geq M_t$) for the same period per area (km^2) and per month, released by the last report as the background. Observed seismicity over the last month are shown as red solid circles. The ROC curve is constructed by iterating through all forecasted earthquake probability density values in descending order. For each probability density threshold, the true positive rate (TPR) and false positive rate (FPR) of the grid cells where actual earthquakes occurred are calculated. In this case, the ROC curve is built based on the probability density distribution of forecasted earthquakes ($M \geq M_t$) per area, as published in the last monthly forecast report. This process involves comparing the forecasted probability density with the actual earthquakes that occurred during that period to evaluate the performance of the prediction model.

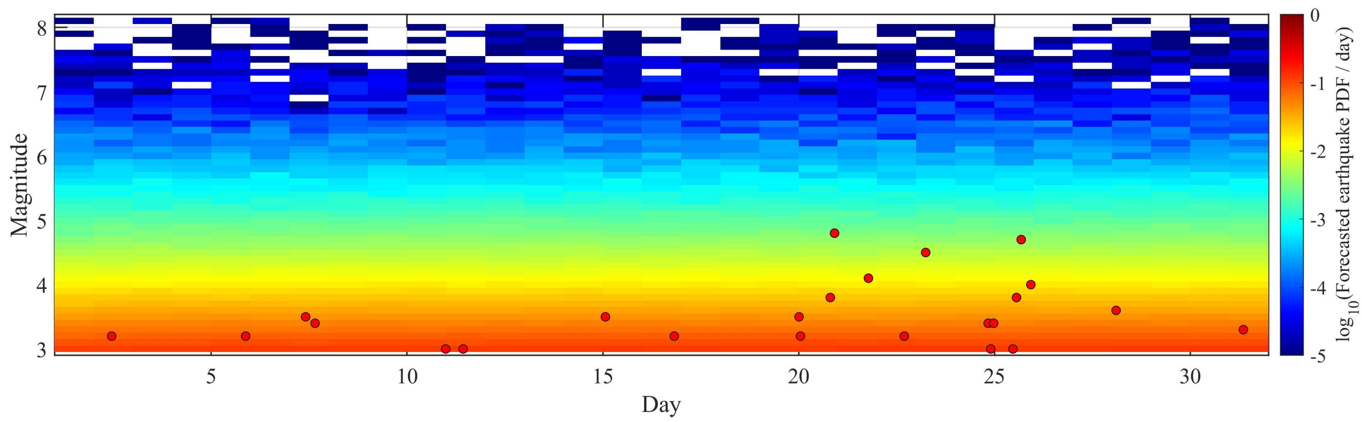


Figure 8. Observed seismicity (red solid circle) over the last month, with the daily probability density distribution (PDF) of the numbers of forecasted earthquakes for the same period, released by last report as the background.

3. Background Information

Since the inception of seismic records at the CSES (~ 780 BCE), there have been 210 earthquakes of *M* 6+, 43 of *M* 7+, and 2 of *M* 8+. Since 1970, there have been 539 earthquakes of *M* 5+, 91 of *M* 6+, 14 of *M* 7+, and 1 of *M* 8+.

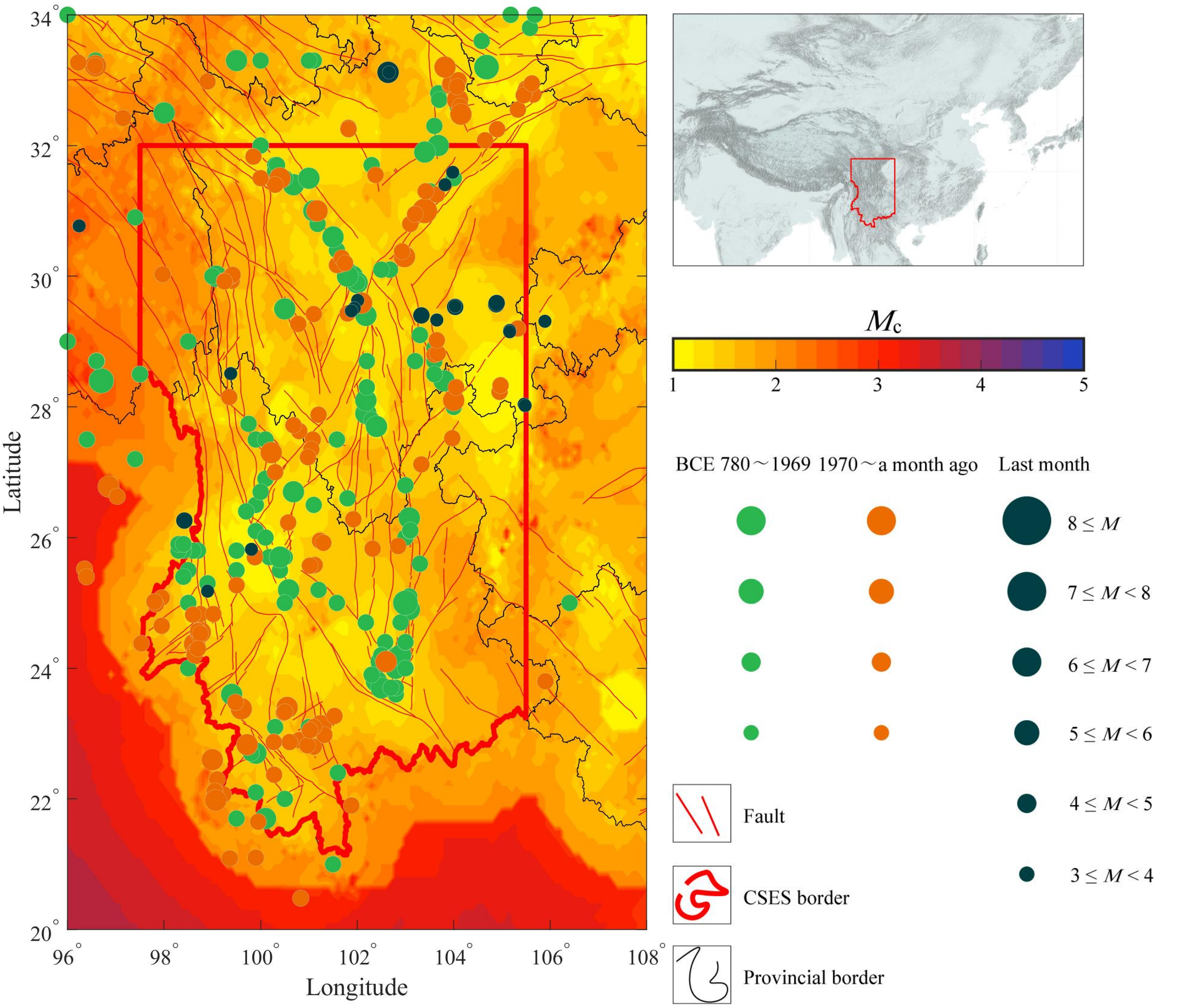


Figure 9. Spatial distribution of earthquakes and faults in the CSES located in the Sichuan-Yunnan region. The background map, showing the completeness magnitude (M_c), is adapted from Li et al. (2024).

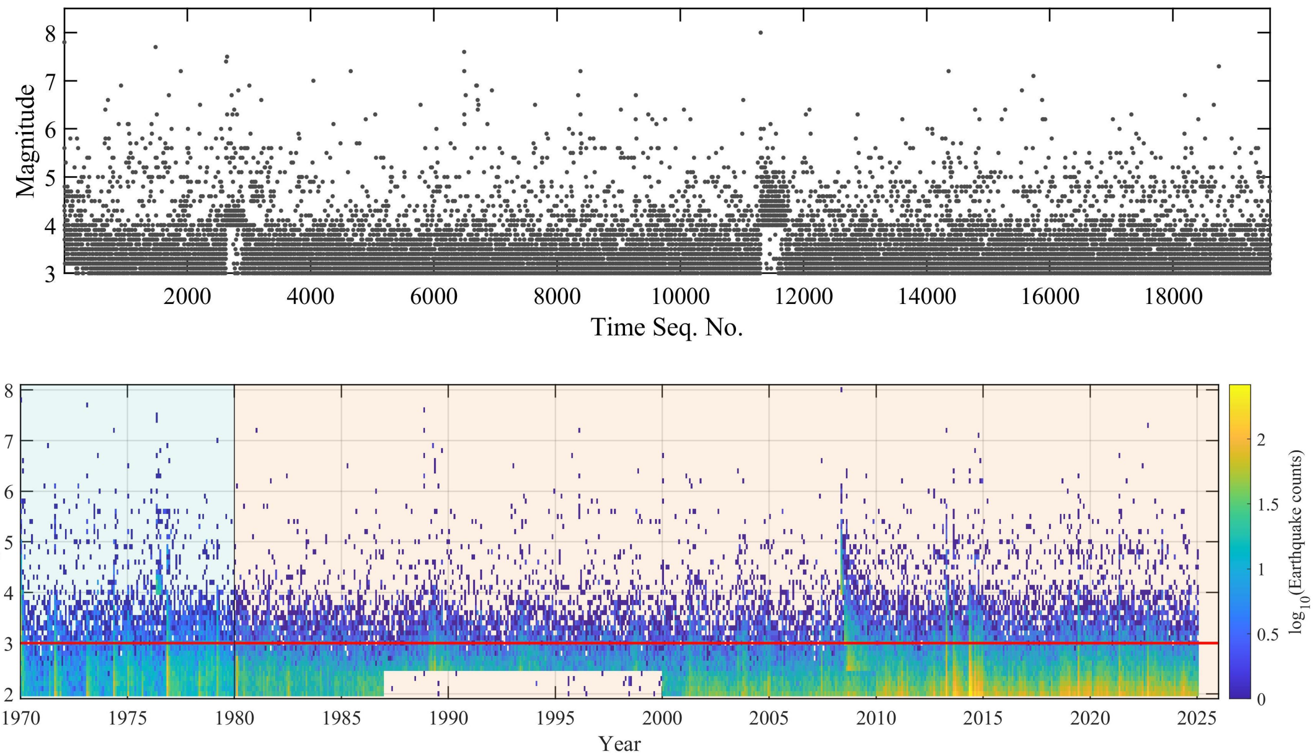


Figure 10. (Top) Magnitude-time sequence plot and (bottom) monthly earthquake counts per 0.1 magnitude bin in the CSES since 1970. The red line indicates the cut-off magnitude M_{co} = 3.0. The light red and light blue shaded regions represent the primary period and the auxiliary time band, respectively. The parameters M_{co}, primary period, and auxiliary period are all utilized for the ETAS model calibration and simulations to build the forecasts.

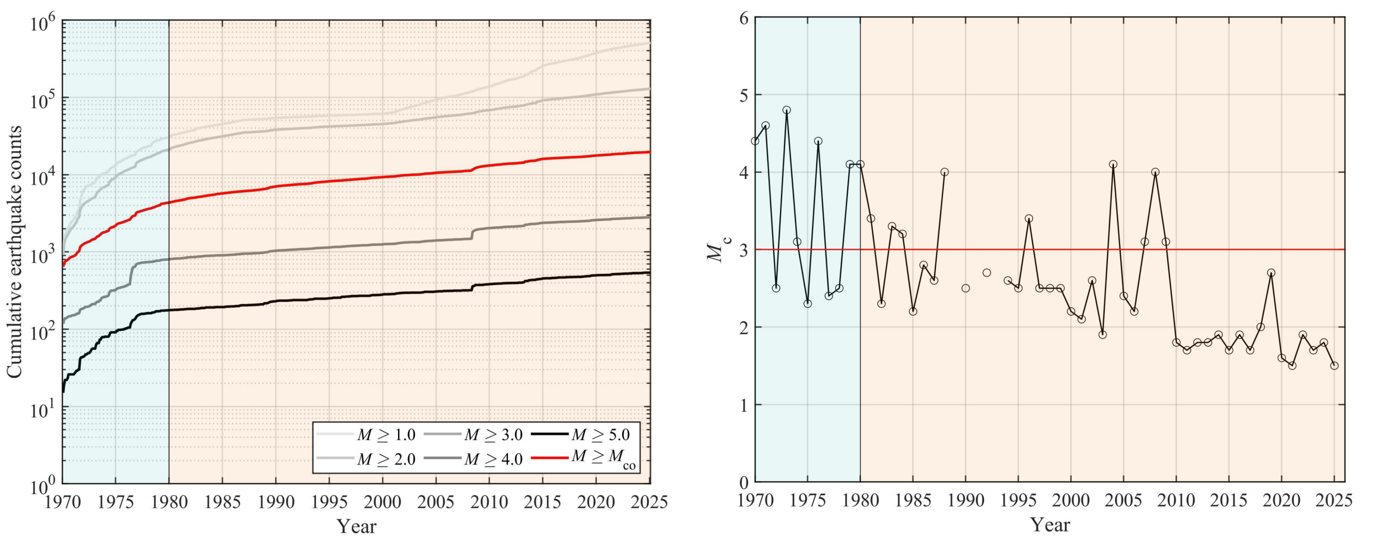


Figure 11. (left) Time series depicting cumulative earthquake counts for different cut-off magnitudes (M_{co} = 1, 2, 3, 4, and 5) in the CSES; (right) Evolution of the completeness magnitude (M_c) over time. The red line indicates the cut-off magnitude M_{co} = 3.0. The light red and light blue shaded regions represent the primary period and the auxiliary time band, respectively. The method to estimate M_c is described in Li et al. (2024).

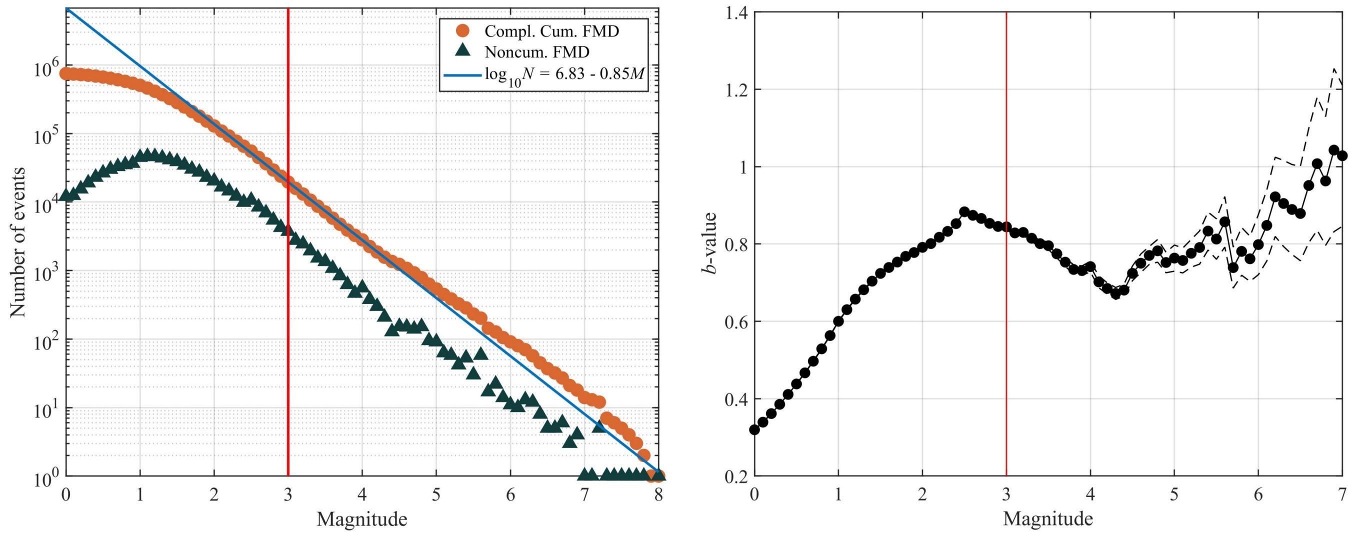


Figure 12. (left) Complementary cumulative and density frequency-magnitude distribution alongside the Gutenberg-Richter (GR) law fitted using earthquakes with magnitudes larger than $M_{co} = 3.0$ for the CSES catalogue since 1970; (right) Variation of the b-value of the GR law as a function of M_{co}, with the thin dashed line indicating the $\pm 1\sigma$ standard deviation. The methods to estimate b-value and to fit the GR law are described in Li et al. (2024).

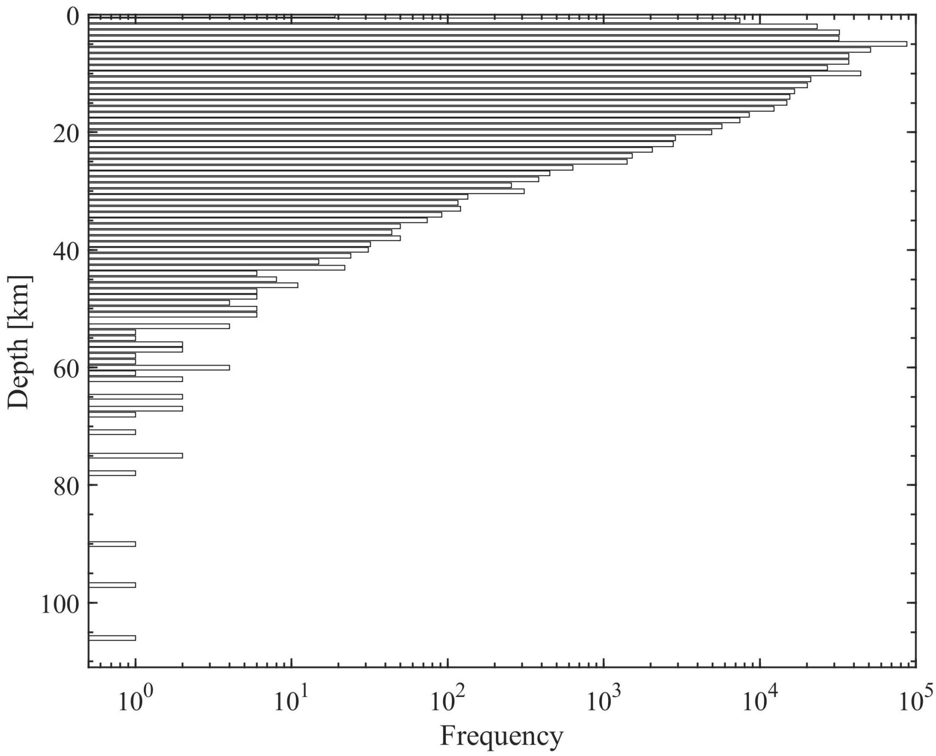


Figure 13. Depth-frequency distribution of earthquakes in CSES.

Acknowledgement

Since the first release of the AoyuX Earthquake Forecasting Series Report on June 4, 2024, we have received many constructive comments and suggestions from colleagues. These colleagues, in the order their feedback was received, include Prof. Jiancang Zhuang, Prof. Zhongliang Wu, Associate Prof. Yu Feng, Prof. Guixi Yi, Prof. Changsheng Jiang, Prof. Yongxian Zhang and Associate Prof. Ke Wu. We express sincere gratitude to all of them! Most of these comments and suggestions have been reflected in the subsequently released series reports.

The earthquake catalogs in this report are sourced from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at the China Earthquake Administration, available at http://10.5.160.18/console/index.action (last accessed: December 18, 2023), with a Digital Object Identifier (DOI) of 10.11998/SeisDmc/SN. The Bayesian Magnitude Completeness (BMC) method used for calculating the spatial distribution of M_c was provided by Dr. Arnaud Mignan. The Epidemic Type Aftershock Sequence (ETAS) model code for simulating earthquake catalogs and calculating earthquake probabilities was derived from a code shared by Dr. Shyam Nandan, which is used in (Nandan et al., 2021a;b; 2022). Fault data were obtained from the GMT Chinese Community at https://docs.gmt-china.org/latest/dataset-CN/CN-faults/ (last accessed: May 31, 2024). This work is partially supported by the Guangdong Basic and Applied Basic Research Foundation (Grant No. 2024A1515011568), and the Center for Computational Science and Engineering at Southern University of Science and Technology.

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Attachment: About AoyuX

AoyuX is a developing online research platform dedicated to advancing earthquake predictability in China. It is produced by the Institute of Risk Analysis, Prediction, and Management (RisksX) at the Southern University of Science and Technology (SUSTech), in collaboration with multiple research institutions. In Chinese mythology, "Aoyu" is a legendary creature with a dragon's head and a fish's body, believed by ancient Chinese to cause earthquakes. The "X" represents the unknown and infinite in Western culture, symbolizing goals and hopes.

The AoyuX platform aims to use cutting-edge statistical seismology tools for forward earthquake forecasting research. It will also serve as an online research platform for forward earthquake forecasting competitions with other models. In the future, the AoyuX platform will integrate geophysical observation data, artificial intelligence (AI), and other technologies to conduct various types of earthquake forecasting research, including statistical, numerical, and physical methods. AoyuX is dedicated to help building a modern earthquake forecasting evaluation system in China, encompassing mid-term, short-term and imminent forecasts. It will develop a new visualization platform for earthquake predictability research, facilitating communication between experts and the public, and serving the combined efforts of experts and the public in tracking and assessing the imminent occurrence of major earthquakes, as well as in mid-term and short-term earthquake forecasting.

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